

P08-R2 / 01 ZASILANIE CZUJNIKA I PRZEKAZNIK

R1 232k 1%: okolo 117 mA typ.; z rownan TI okolo 99..139 mA z tolerancja rezystora. Nie jest to limit 20 mA.

K1: COM 3/6, NO 4/5. Rozlacza +5V i powrot; AGND_SENSOR nie laczyć z GND poza K1.

U8 wymusza TPS_EN=0 przy brownout 3V3. R15/R16 zmniejszaja EN ponizej 0.66 V gdy 3V3 <1 V.

U8 zwalnia do 700 ms; przed pierwszym wlaczeniem odczekac 750 ms od SENSOR_OK=1. K1 moze zalaczyc wczesniej niz U1.

Format S1 1/3 slot S1 level 5; schematic for review, hardware not tested

Prototype / schematic review, no PCB

Sheet: /
File: P08.kicad_sch

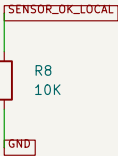
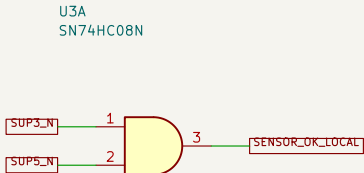
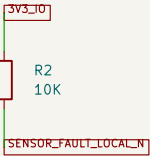
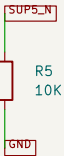
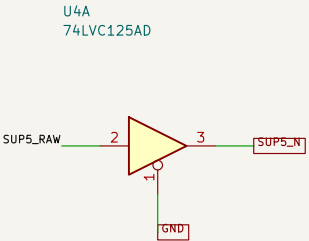
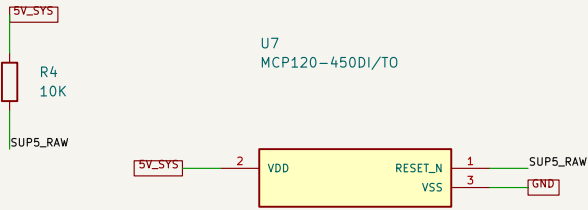
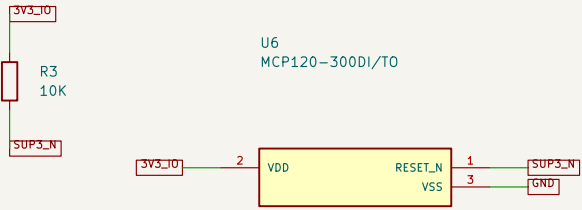
Title: **EGRLab P08 SENSOR**

Size: A3 Date: 2026-10-05
KiCad E.D.A. 10.0.6

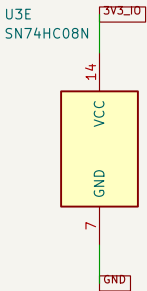
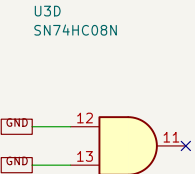
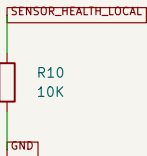
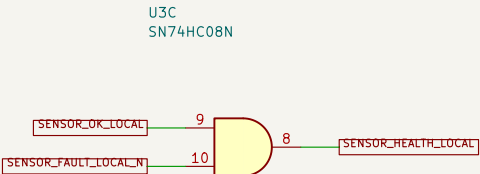
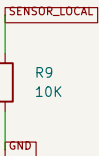
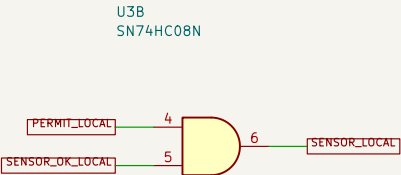
Rev: **P08-R2**
Id: 1/4

P08-R2 / 02 GOTOWOSC I WARUNKI ZEZWOLENIA

SENSOR_OK_LOCAL = SUP3_N & SUP5_N; niezależny od PERMIT i od napiecia za przekaźnikiem.



SENSOR_LOCAL = PERMIT_LOCAL & SENSOR_OK_LOCAL. HEALTH_LOCAL = SENSOR_OK_LOCAL & FAULT_N.



Brak sprzętowego latch FAULT. U1 ogranicza prąd; CORE musi zatrzasnąć błąd i cofnąć PERMIT.

Format S1 1/3 slot S1 level 5; schematic for review, hardware not tested

Prototype / schematic review, no PCB

Sheet: /LOGIC/
File: LOGIC.kicad_sch

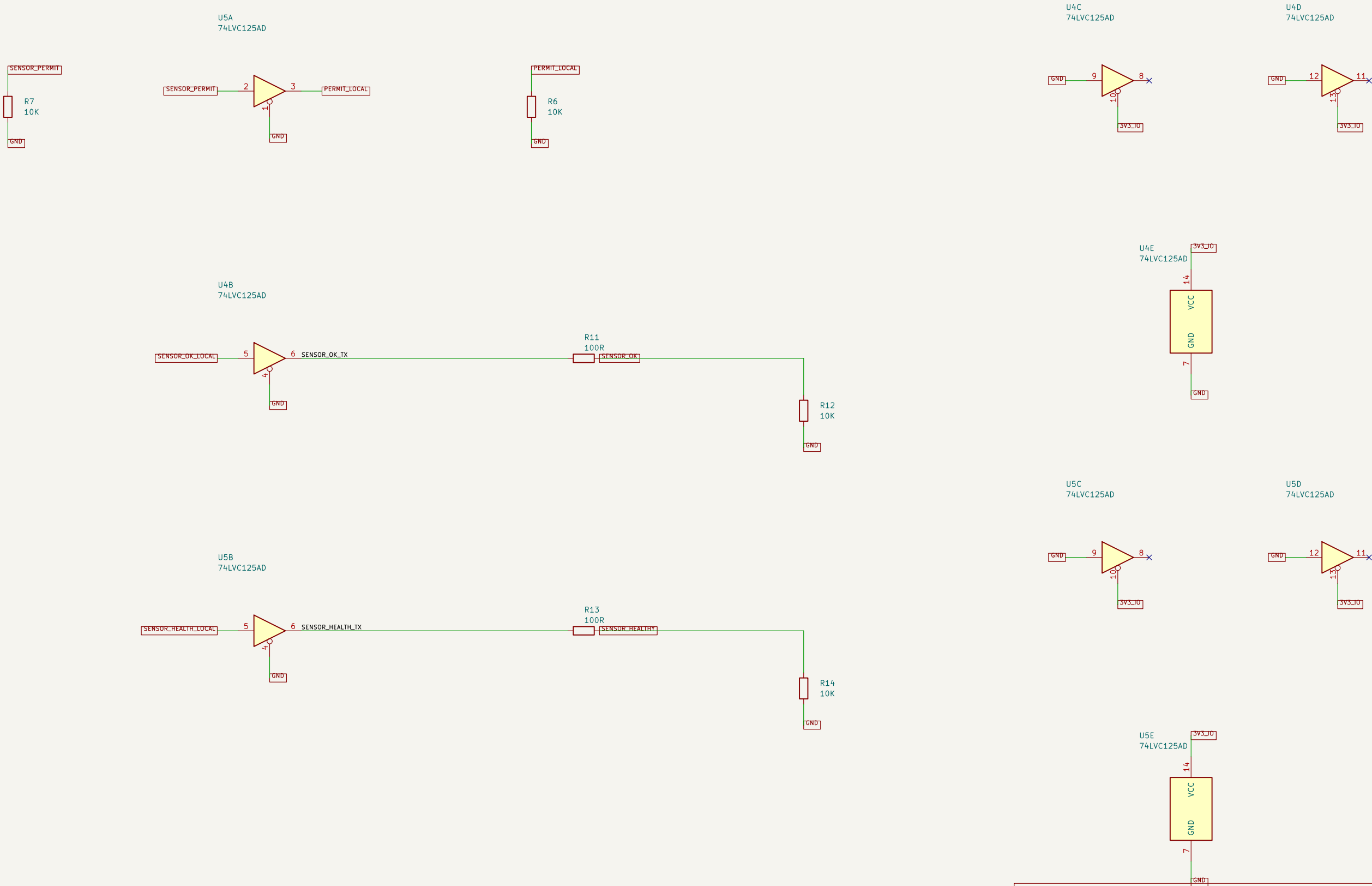
Title: EGRLab P08 SENSOR

Size: A3 Date: 2026-10-05
KiCad E.D.A. 10.0.6

Rev: P08-R2
Id: 2/4

P08-R2 / 03 BUFORY SYGNALOW MIEDZY PLYTKAMI

U4/U5: Nexperia 74LVC125AD z loff, bez zamiany na HC125. Zasilanie obu 3V3_I0.



Wyjścia SENSOR_OK i SENSOR_HEALTHY: 100R szeregowo, 10k do GND. Brak zasilania -> LOW u odbiorcy.

HEALTHY=1 nie dowodzi zamknięcia K1, poprawnego mapowania pinów EGR ani obecności 5 V na czujniku.

Format S1 1/3 slot S1 level 5; schematic for review, hardware not tested

Prototype / schematic review, no PCB

Sheet: /10/
File: 10.kicad_sch

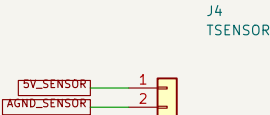
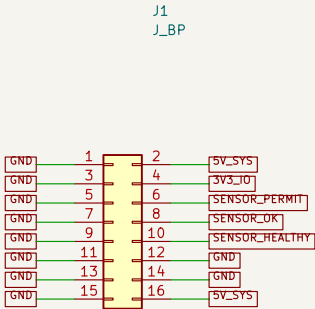
Title: EGRLab P08 SENSOR

Size: A3	Date: 2026-10-05	Rev: P08-R2
KiCad E.D.A. 10.0.6		Id: 3/4

P08-R2 / 04 J_BP, LISTWA SERWISOWA, TSENSOR

J_BP (krawedz A, slot S1 poziomu 5, srodek x=26,5): nieparzyste GND; 2/16 5V_SYS, 4 3V3_I0, 6 SENSOR_PERMIT (z P04), 8 SENSOR_OK (do P04), 10 SENSOR_HEALTHY (do P03), 12/14 GND (zapas masy).

Zastepuje wiazki R1: LV08 (P02-R3/J8), SENSOR (P04-R2.1/J4), SFAULT (P03-R2/J6). Nazwy sieci jak w R1 i P04-R2.2: P12 laczy po nazwie.



J4 TSENSOR: pole przewodow 2 x AWG22 z kotwa przy krawedzi x=0 (strona panelu), para do portu TEST na panelu. AGND_SENSOR nie laczyc z GND poza K1.



SERWIS (krawedz B, x=10..43 w slocie): GND na pinach 1 i 13; kolki przez R19..R29 przy wezlach (1k, TPS_EN 10k); pin 1 rezystora = siec, pin 2 = SRV_<siec>.

5V_SENSOR/AGND_SENSOR mierzyc roznicowo na porcie TEST (para J4), nie na listwie: zsunieta sonda nie moze ominac styku powrotu K1.

TEST: mapowanie 5V/GND czujnika zatwierdzone poza P08. LOGGER: SENSOR_PERMIT=0; zasilanie czujnika pochodzi z ECU.

Format S1 1/3 slot S1 level 5; schematic for review, hardware not tested

Prototype / schematic review, no PCB

Sheet: /CONNECT/
File: CONNECT.kicad_sch

Title: EGRLab P08 SENSOR

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-10-05

Rev: P08-R2

Id: 4/4